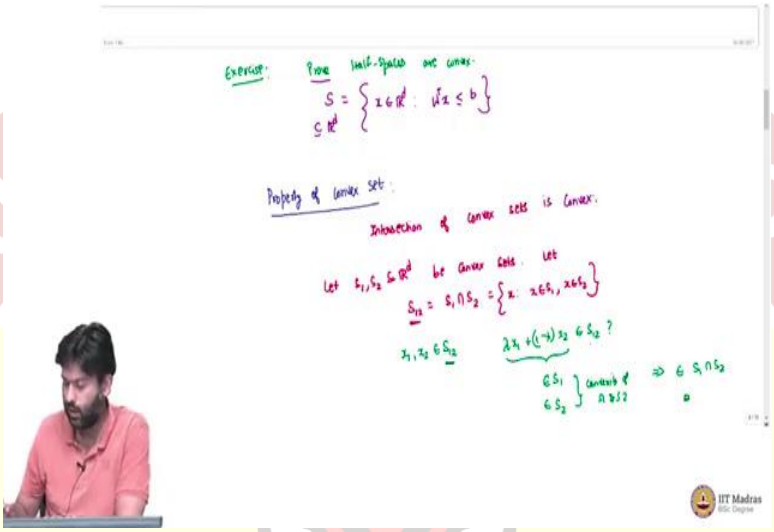


IIT Madras

ONLINE DEGREE

Professor Arun Rajkumar
Department of Computer Science and Engineering
Indian Institute of Technology, Madras
Properties of Convex Sets

(Refer Slide Time: 00:15)



Now, we are going to see an interesting property of convex sets. There are several properties I mean it would be too much to list down all properties in this course but we will see some properties which are relevant to our discussion. So, we look at one property in particular, this is a property of convex sets.

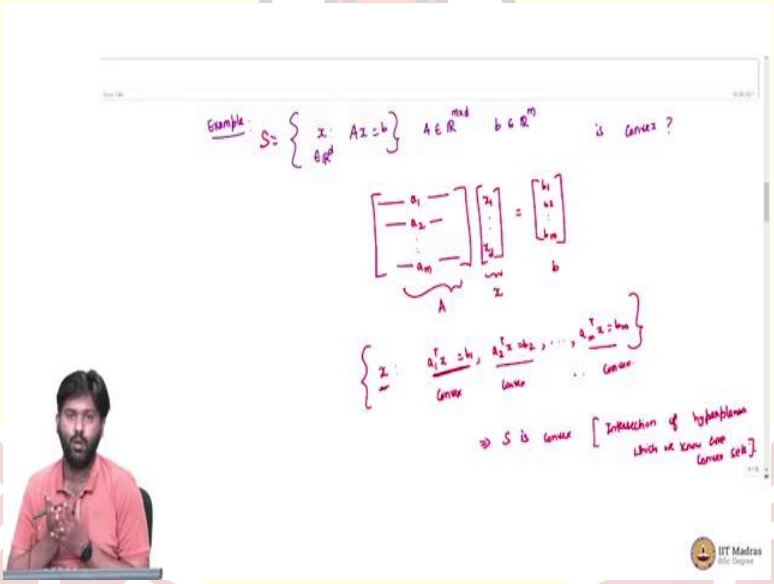
The property is the following; Intersection of convex sets; what can we say about the intersection of convex sets? Let us say I give you two convex s1 and s2 and I ask can you say something about the intersection of these two sets. So, let s1 , s2 think of this again a subset of some Rd the convex sets sometimes I might drop the word sets when I call s1 and s2 subset of r d s convex means that they are sets.

Now let us, let me call s1 2 as simply s1 intersection s2 which is simply the set of all x such that x belongs to s1 , x belongs to s2, usual definition of intersection; is s1 2 convex? What do you think? Well, let us see; Now if I had to show s1 2 is convex how would I go about doing that? Well, I have to pick some x1 and x2 let us pick two points x1 and x2 belonging to s1 2 and then I want to show λ x1 + 1 - λ x2 should also belong to s1 2, is this true?

Well, what can I say about λ x1 + 1 - λ x2, well it belongs to s1 simply because x1 and x2 belongs to s1, why do they belong to x1? Because they belong to the intersection so they have to belong to both the sets which means they belong to s1 and λ x1 + 1 - λ x2 also belongs to s2. Now this is because of convexity of x1 and x2. So, that will force these two this point belong to both s1 and s2 now which implies that it belongs to the s1 intersection s2 which is, which is what we want.

So that is the proof, so this, so I took some point which is a convex combination of the two points x1 and x2 and then I argued that it belongs to s1 it belongs to both the sets and so also the intersection. So, the conclusion is that intersection of convex sets is convex, it was quite easy to see this. Well, it can be used in a lot of cases where it may not be very obvious to prove a set is convex but then if you can show that set is an intersection of simpler sets which are convex sets then you can also show that the set you care about is convex.

(Refer Slide Time: 03:27)



Some examples, let us say where this can directly be useful? Let us say we look at the set of all x such that A x equals b for some A let us say m cross d and b which is in m, set of all x of course is in Rd you have a matrix A which is m cross d and vector b which is a R m and I am just looking at the set of all solutions to the equation Ax equals b.

Now is this convex?, so is it convex; well, you can use the definition but here is another simpler way to see this not necessarily the only way but this is one way. So, now what does Ax equal to b mean? Well, I have think of this as the rows of A as A1, A2 dot-dot-dot and then there are going

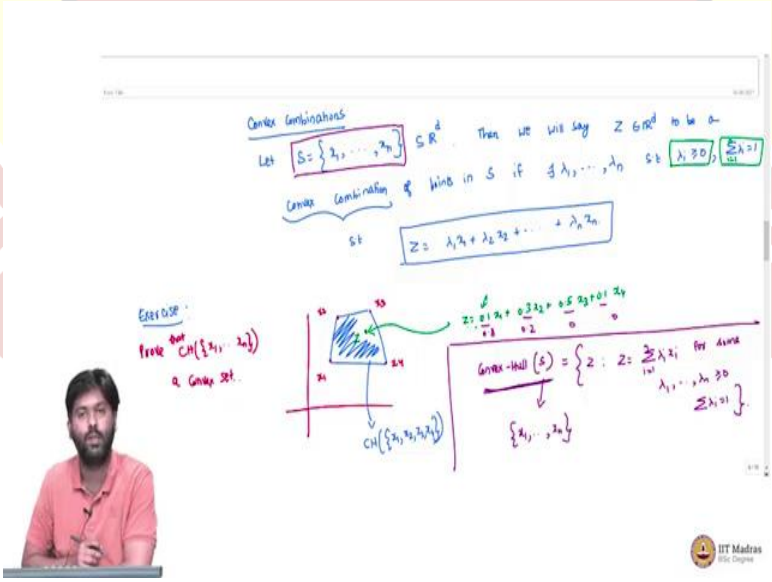
to be m such rows and then you have a vector x which satisfies ax equals v1, v2 dot-dot-dot vm so this is my b this whole thing is my A, this is I can even write this as let us say x1 dot-dot-dot x d this is my x.

Now, which means what the set of x that satisfies this equation is exactly the set such that A1 transpose x is b 1, A2 transpose x is b 2 dot-dot-dot Am transpose x is b m so why; because, because it has to satisfy a x equals b whereas the matrix now each of these rows have to necessarily satisfy the individual equations.

Well, we already know the set of all x such that a 1 transpose x equals b 1 is convex, so this is a convex set, this is a convex set and so on this is convex set. Now we are asking for the set of all x which satisfy all of these together and we just showed that intersection of convex set is convex so which implies the set a let me call this s, so s is convex this is just an intersection of hyper planes which we know are convex sets.

So, it is a simple proof you can also go over the actual definition you would get the same thing, so yeah good it is good to know that solutions to linear systems of equations are actually convex sets. Fine let us move on.

(Refer Slide Time: 06:31)



Now, let me define something for you, now we would often look at things what we will call as convex combinations. So, what is a convex combination? Well, let us look at a set s which is not,

which is just a bunch of points let us call these x1 to xn all of these is in Rd you have a bunch of points in d dimensional space.

Then we will say a new point z which is also in Rd to be a convex combination of points in s if there exists λ 1 dot-dot-dot λ n such that λ i's are all greater than 0 and sum over λ i equal to 1 such that your z equals λ 1 x1 + λ 2 x2 dot-dot-dot λ n x n.

All we are saying here is that I have a bunch of points instead of 2 now I have a bunch of points x1 to x n and I am saying in earlier we were looking at line segments joining these two points, now I am saying that a point new point z if you can get it as a combination in particular a convex combination of the points in s if it can be written in this particular form.

So, λ 1 x1 + λ 2 x2 + dot-dot-dot λ n x n; what does that mean? Well, if I had let us say again we are in r 2 and let us say I have a bunch of points let us call this x1, x2, x3, x4 and now maybe here is a point, now I will call the green point to be a convex combination of the red points. If this point may be z is 0.1 times x1 + 0.3 times x2 + 0.5 times x 3 + 0.1 times x4 I mean I may not get the exact numbers here corresponding to this geometry of this set but does not really matter.

So, the main thing is that I can express z as some combination of x1 to x n where the combinations themselves are positive and they sum up to 1 they have to sum up to 1. So, this is i equals1 to n, they can be 0 also. There is nothing in the definition that stops one of these guys to be 0 for example so you can have that if you want. Now why are, why is this useful?

Well this is useful because of our second definition which is what we call as the convex hull of a set of points s, so again think of s as simply a bunch of points x1 to x n now we are going to think of we are going to define a new set which is which we will call as the convex hull of s as simply the set of all z such that z equals sum over i equal to 1 to n λ i xi for some λ 1 to λ n greater than or equal to 0 sum over λ equals1.

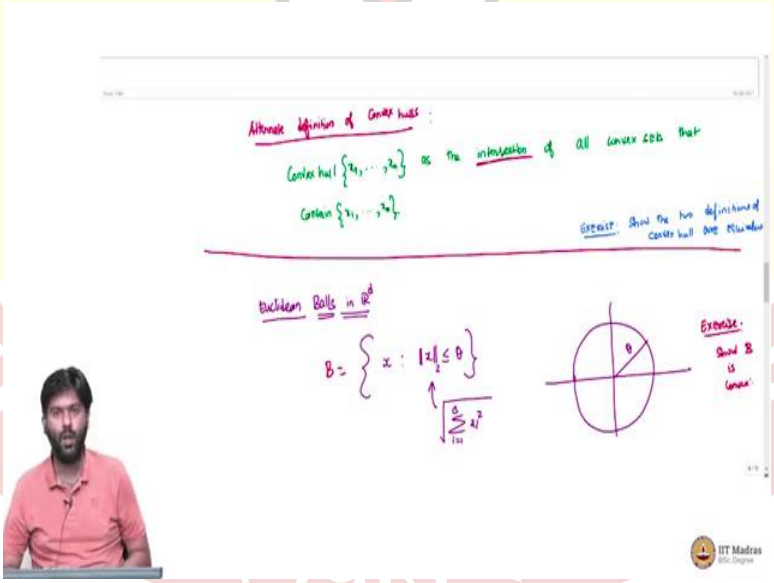
Essentially, we are saying the convex hull of a set s is just the bunch of all points which can be got as convex combinations of the original set of points that you started off with, so that is the definition. So, you can use any λ in this case we use 0.1, 0.3, 0.5, 0.1 to get a point maybe you will use point 0.8, 0.200 to get a different point so in this space.

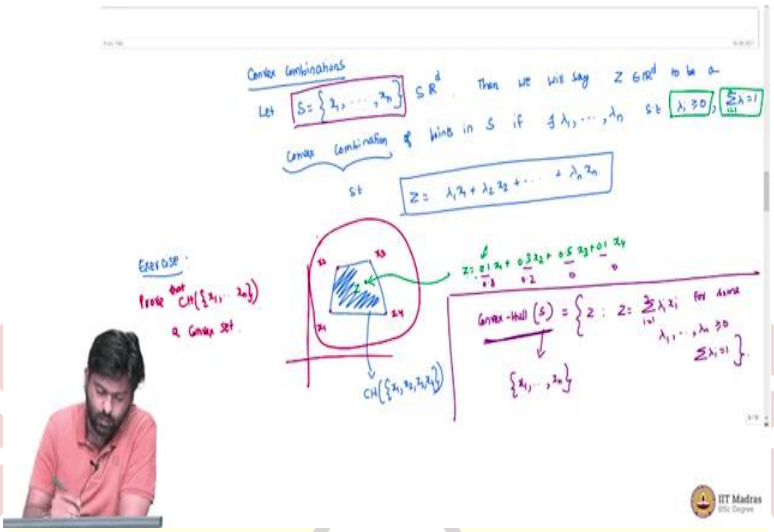
Now I collect all such points and I call the set that I generate in that fashion as the convex hull of the original set s where s itself is again x1 to x n there is just a bunch of finite number of points. Now, okay so in this particular example again let us think of the same set of points x1, x2, x3, x4 now where do you think is the convex hull of s? In other words where are the set of points which can be expressed as some convex combination of x1, x2, x 3 and x4 so if you are not already seeing it maybe you should think pause a bit and think about this.

So, if you actually compute the convex hull of the set it is simply going to look like this, the exact set itself would be this. So, it would be all the points inside this, this is the convex hull sometimes abbreviated as ch of x1, x2, x3 and x4. Well, it is a convex hull which means that this set itself must be a convex set.

Now here is an exercise for you to think about and it is not too hard; is c h of x1 to x n a convex set or so rather it is a convex set because we are calling it the convex set it better be a convex set I will say prove that ch of x1 to x n is a convex set, this is a simple exercise to do, okay.

(Refer Slide Time: 12:55)





Here is another way to think about convex hulls, alternate definition of convex hulls, we would not define too formally I just want to give you flavor, so now you can think of a convex hull of a set of points as the intersection of all convex sets that contain x1 to x n. You take any set that contains x1 to x n, so there may be several sets that will contain x1 to x n now take all sets which are convex sets which contain x1 to x n.

When I say contain x1 to x n in the previous example it should be clear that well here is one set which is convex and contains x1, x2, x3 and x4. So, there may be infinitely many sets that contain x1 to x n. Now the claim is that if you intersect all these sets you again get a convex set that is by property of intersection of convex sets is a convex set.

Now, the claim is that this set is indeed the convex hull of x1 to xn. So, this is another characterization of the convex hull which might be useful in some cases where if you want to define something as a (con) if you want to prove something is a convex set one way to again argue is that show that it is a intersection of any set that contains a set of points if the original set is convex then the intersection of all these convex sets will be the convex hull. So, I just wanted to also mention that.

Now, it is again an exercise may be slightly harder than the previous exercise we have been seeing. So, show the two definitions of convex hulls are equivalent, So, how do you show two definitions are equivalent? Well, you take a point which satisfies the first definition and argue that it should also satisfy the second definition and you do the other way also.

So that is one way to show that two sets are exactly the same, if one is contained in the other and the other is contained in the first then these two sets are exactly the same. So, that may be one way to show this but I will let you think about this, so this is an exercise. So, what are other examples of convex sets?

Again, a lot of these examples are very easy to show that they are convex and that is why I am encouraging you to do them as exercises instead of doing a lot of I mean applying the definition again and again but I think it should be, it should be a very useful exercise for you to get a grip of convexity by trying out these exercises that I am suggesting, okay here is one more example of a convex set like balls.

So, what are balls in let say Rd? These are let us Euclidean balls, so set of all x such that you know you can think of it as the norm of x is less than or equal to some theta, so basically in us by norm it is a Euclidean norm by which I mean sum over i equals one to d xi squared and the square root. So, this has to be less than or equal to theta of course this is simply the units I mean not the unit circle, the circle with the radius theta.

So, this would be your ball and it is very clear at least visually that this is convex but then again it is a very useful exercise to try this out. So, show let us call this b, b is convex. Again, a simple example but nevertheless good exercise to do it once.